



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: BCA/BCA AI & DS

Semester: 4

Name of the Paper: Fundamentals of Data Analytics

Paper Code: TBC411/TBD 402

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Define Data Science. Explain the types of digital data and discuss the key characteristics of data with suitable examples. CO1

OR

b. Describe the data lifecycle in conventional data systems. What are the major challenges faced by traditional systems in handling large-scale data? CO1

Q2. (10 Marks)

a. Explain the evolution of Big Data. Define Big Data and discuss the 3Vs of Big Data along with their significance. CO1

OR

b. What are the challenges associated with Big Data? Explain the non-definitional traits of Big Data in detail. CO1

Q3. (10 Marks)

a. What is Hadoop? Explain its role in Big Data analytics. Compare RDBMS and NoSQL databases with suitable examples. CO2

OR

b. Explain Exploratory Data Analysis (EDA). Discuss the steps involved in data preparation and data summarization with examples. CO2

Q4. (10 Marks)

a. Describe data distribution and explain how skewness is measured. Illustrate with suitable diagrams. CO3

OR

b. Explain mean, variance, and standard deviation. Why are they important in data analysis? CO3

Q5. (10 Marks)

a. What is statistical inference? Explain the frequency-based approach with examples. CO3

OR

b. Describe hypothesis testing. Explain the methods of testing hypotheses using confidence intervals and p-values with examples. CO3